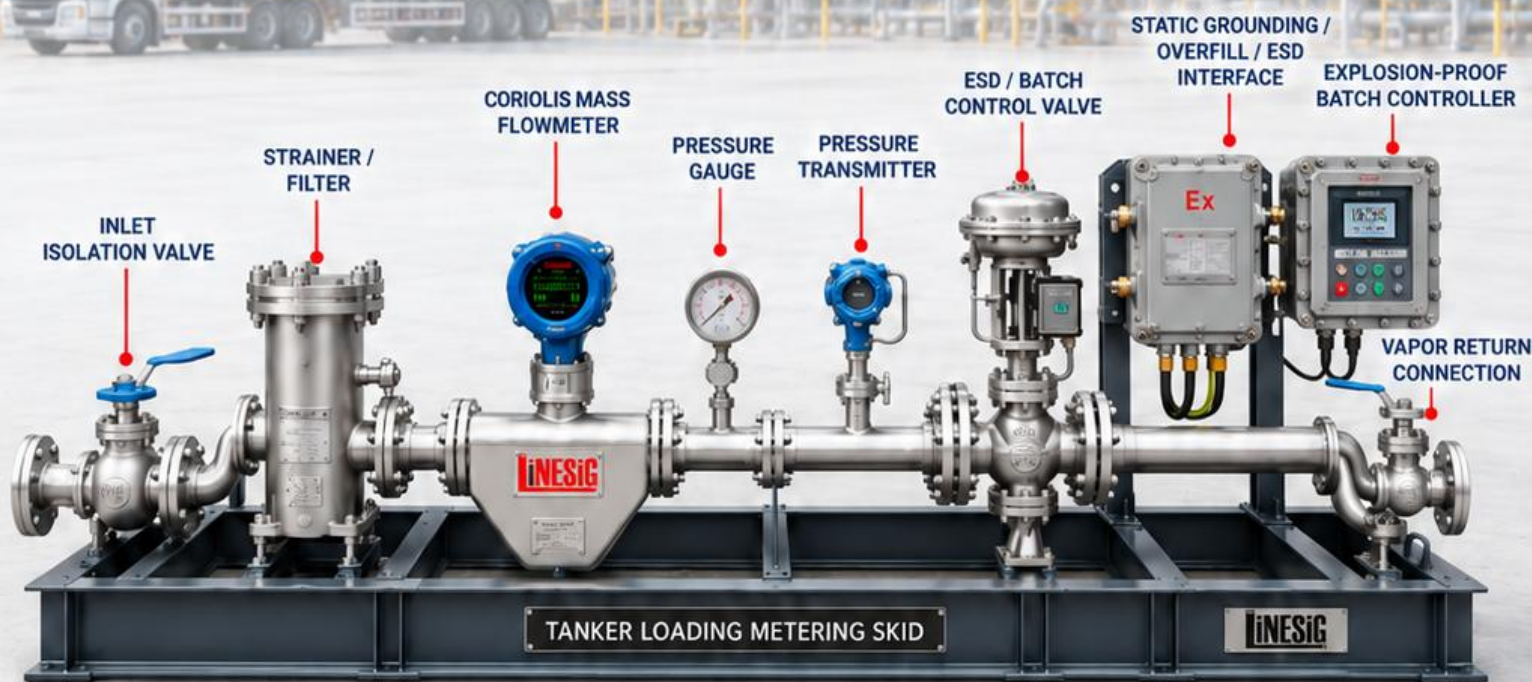




LNS-MS SERIES

Tanker Loading Metering Skids

TECHNICAL DATASHEET: **LNS-MS-LPG**



Product Model: **LNS-MS-LPG**

Application: Liquefied Petroleum Gas

1. Product Overview

The **LINESIG LNS-MS-LPG** Series is a modular family of skid-mounted tanker loading metering systems developed for controlled and traceable transfer of liquid hydrocarbon products from terminal storage or transfer headers to road, rail or marine loading interfaces. The standard **LNS-MS-LPG** architecture is based on Coriolis mass flow measurement wherever technically suitable, together with local explosion-proof batch control, flow regulation, pressure monitoring, Static grounding system, Pressure and temperature transmitter and other loading safety interfaces.

Design philosophy:

The product family is configurable. Final size, flow range, pressure class, materials, meter model, batch controller certification and loading interface are selected against the actual project datasheet and hazardous-area requirements.

Product Family Matrix:

Model	Primary Service	Preferred Metering	Special Design Considerations
LNS-MS-LH	Refined & light liquid hydrocarbons	Coriolis mass flowmeter	Custody/transaction measurement, loading arm interface, vapour return where required
LNS-MS-FO	Fuel oil, base oil & lube oil	Coriolis preferred; PD alternative where justified	Viscosity, heating/tracing, pressure drop, temperature compensation
LNS-MS-LPG	Liquefied petroleum gas	Coriolis mass flowmeter	Pressurised liquid service, vapour return/equalisation, ESD, anti-flash hydraulic design
LNS-MS-LNG	Liquefied natural gas	Cryogenic Coriolis mass flowmeter	Cryogenic materials, cooldown, vapour/BOG interface, ESD and thermal contraction

2. LNS-MS-LPG - Liquid Petroleum Gas Skid

For Liquid Petroleum gas, the configuration emphasizes accurate batch measurement, clean flow conditioning and loading safety.

Design Aspect	LNS-MS-LPG Basis
Meter	Coriolis mass flowmeter suitable for pressurized LPG
Hydraulic requirement	Maintain adequate liquid pressure through skid to prevent flashing
Vapour return	Dedicated vapour return/equalization interface where required
Filtration	Product-specific strainer/filter; Jet A1 cleanliness requirements require dedicated review
FCV	Pneumatic or electro-pneumatic flow/batch control valve
Batch controller	Explosion-proof local controller or TAS-integrated controller
Loading arrangement	Top or bottom loading according to project requirement
ESD	Fail-safe ESD isolation and batch shutdown philosophy
Pressure/temperature	Continuous monitoring as required for operational and transactional calculations
Breakaway / ERC	Available at loading interface where project requires
Grounding / overfill	Integrated loading permissive
Hazardous area	Explosion-protected equipment according to project classification

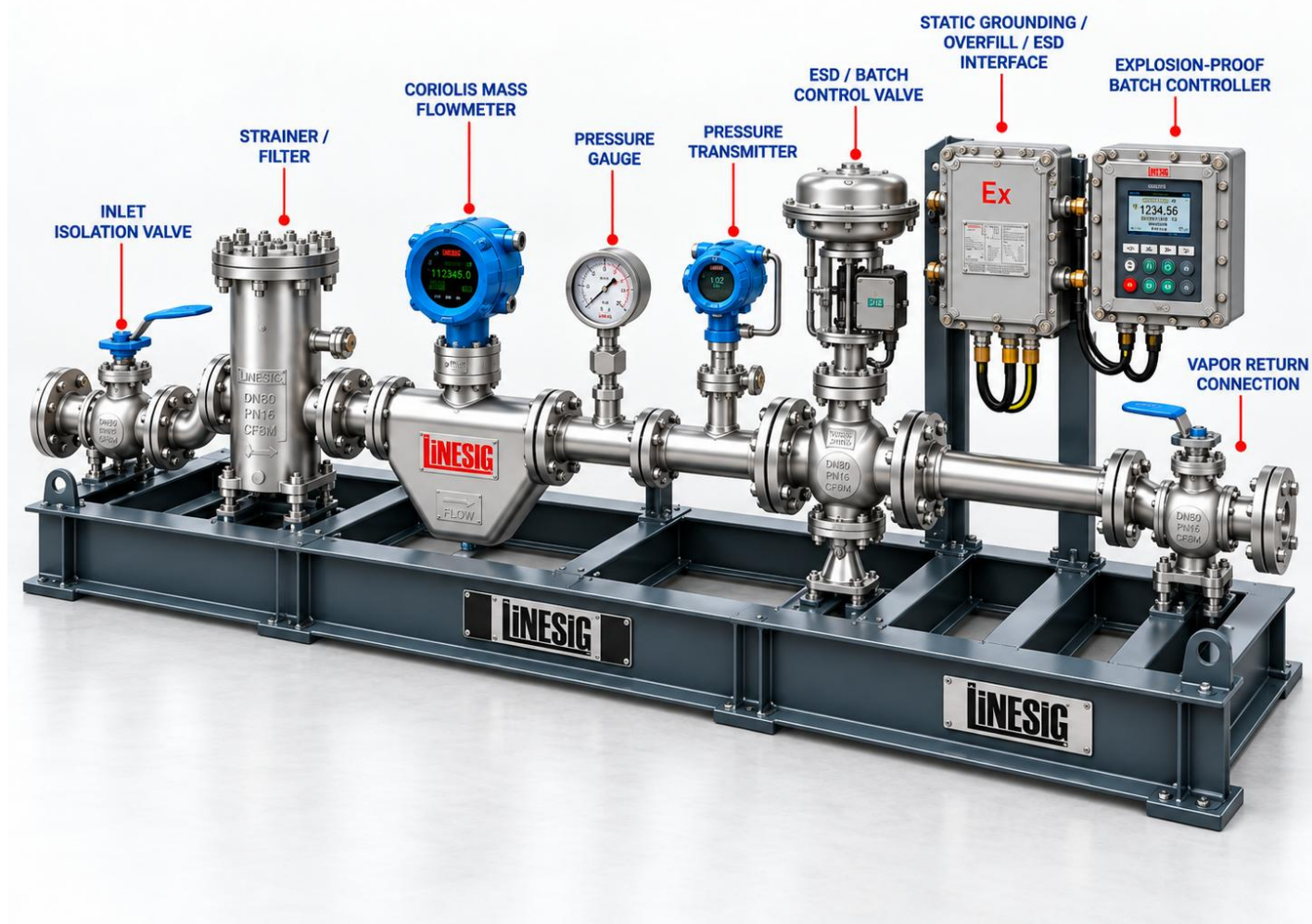
Standard Process Sequence:

Inlet isolation → Strainer / Filter → Air Eliminator or Gas Separator where applicable → Pressure / Temperature Measurement → Coriolis Mass Flowmeter → FCV / Batch Control Valve → Check / Isolation → Loading Arm or Hose Interface-vapour recovery where applicable.

Functions:

Explosion-proof batch controller coordinates preset quantity, start/stop logic, meter totalization and batch termination. Grounding, overfill, ESD and loading-arm permissive can be hardwired into the batch logic or communicated to TAS/PLC/SCADA.

Air elimination is applied where entrained gas can affect liquid metering. For LPG/LNG, gas handling philosophy is product-specific and should not be copied directly from atmospheric liquid service.



3. General Technical Specification

Parameter	Standard / Configurable Requirement
Product family	LNS-MS Series - tanker loading metering skids
Model	LNS-MS-LPG Type- Liquid Petroleum Gas
Metering principle	Coriolis mass flowmeter preferred where applicable
Flow range	Project-specific; meter and line size selected from minimum / normal / maximum loading rate
Line size	Project-specific
Design pressure	Project-specific
Design temperature	Project-specific
Measurement outputs	Mass flow, mass total; volumetric/density/temperature outputs where supported by selected Coriolis meter
Meter accuracy	Vendor/model/project-specific; custody-transfer configuration where required
Repeatability	Vendor/model/project-specific
Pressure drop	Calculated for complete skid at minimum/normal/maximum flow and product viscosity
Process connection	ASME / EN / project flange standard
Piping material	Carbon steel / stainless steel / cryogenic stainless steel / project-specific
Skid frame	Structural steel base frame with lifting points; finish per project coating specification
Hazardous area	ATEX / IECEx / NEC / local requirement as specified by project
Electrical supply	Project-specific
Instrument air	Required where pneumatic FCV or pneumatic actuators are selected
Communication	Modbus RTU/TCP, Ethernet, pulse, frequency, analog/digital I/O as selected
Control interface	Standalone batch controller, PLC/TAS/SCADA integrated, or hybrid

Why Coriolis is the Preferred LNS-MS-LPG Metering Technology:

- Direct mass measurement without requiring density compensation to derive mass flow.
- High repeatability and strong suitability for batch loading and product accountability.
- Ability to provide density and temperature-related diagnostics where supported by the selected transmitter.
- No rotating measurement element in the primary sensor.
Suitable across a broad range of hydrocarbon and liquefied-gas applications when correctly sized for pressure drop, temperature and two-phase risk.

Selection note:

Coriolis is preferred, not automatically mandatory. Final meter selection must account for flow range, viscosity, pressure drop, line size, custody requirements, product temperature and two-phase/flash behaviour.

4. Standard Scope of Supply

Item	Standard Scope / Description	Status
Skid base frame	Fabricated structural skid with lifting provision and equipment supports	Standard
Inlet isolation valve	Manual or actuated as specified	Standard / configurable
Strainer / filter	Basket, duplex or product-specific filtration arrangement	Standard / configurable
DP indication	Local DP gauge and/or DP transmitter across filter where required	Recommended
Air eliminator / gas separator	Liquid-service air/gas removal where applicable	Service-dependent
Pressure gauge	Local pressure indication	Standard
Pressure transmitter	Remote pressure monitoring / batch or TAS interface	Standard / configurable
Temperature element / transmitter	Product temperature measurement	Service-dependent
Coriolis mass flowmeter	Primary metering device	Preferred standard

Item	Standard Scope / Description	Status
Flow control / batch valve	FCV, preset valve or batch shutoff arrangement	Standard
Check valve	Backflow prevention where required	Recommended
Explosion-proof batch controller	Local display, keypad, preset, totalisation and I/O	Standard for local batch control
Local ESD pushbutton	Emergency stop input to batch/terminal shutdown logic	Recommended
Grounding interface	Tanker static-ground permissive	Recommended for tanker loading
Overfill interface	Independent tanker overfill permissive/trip	Recommended
Junction / marshalling box	Field signal and power termination	As required
Loading arm / hose interface	Mechanical outlet connection to loading system	Interface provision

5. Explosion-Proof Batch Controller Functional Requirements

Function	Requirement
Preset quantity	Operator/TAS entry of target batch quantity
Live indication	Flow rate, batch delivered, target, product/status
Start/stop	Local controlled batch start and stop
Valve control	Open/ramp/close FCV or batch valve according to sequence
Meter input	Mass/volume pulses, frequency, digital or communication interface
Safety permissives	Grounding, overfill, ESD, route/pump permissives
Transaction data	Batch total, start/end time, alarms and operator/order reference
Communication	PLC/TAS/SCADA integration where required
Hazardous area	Certification selected to project classification

6. Instrumentation, Control & Safety Interface

Tag / Function	Typical Instrument / Device	Purpose
PI / PG	Pressure gauge	Local pressure indication
PT	Pressure transmitter	Remote pressure monitoring / alarms
TT / TE	Temperature transmitter/element	Product temperature
CMF	Coriolis mass flowmeter	Mass flow, batch total and supported diagnostics
FCV	Flow control / batch valve	Loading-rate control and batch termination
BC	Explosion-proof batch controller	Preset, totalisation, sequence and I/O
ESD-PB	Emergency stop pushbutton	Immediate loading stop command
GS	Static grounding monitor/interface	Ground permissive before and during transfer
OFP	Overfill protection interface	Independent tanker overfill permissive/trip
LS/PS	Package switches as required	Interlocks / local package protection

Minimum Batch Permissive – Typical:

- Valid product / order / batch selected.
- Correct product route available and associated transfer pump ready.
- Grounding status is healthy for road tanker loading.
- Overfill protection interface healthy where provided.
- Meter and batch controller healthy.
- FCV / batch valve in correct initial position.
- No active ESD or critical skid fault.
- Loading arm / hose connection permissive where instrumented.

Trip / Shutdown Actions – Typical:

- ESD command: close batch/ESD valve and stop pump demand according to project cause-and-effect.
- Grounding loss during tanker loading: controlled or immediate shutdown according to project safety philosophy.
- Overfill trip: stop batch and close transfer valve(s).
- Meter or controller critical fault: inhibit automatic batch completion until safe fallback is defined.
- High/low pressure or abnormal process conditions: alarm and trip where required by design.

7. Materials, Options & Ordering Data

Component	Typical Material / Option
Skid frame	Painted carbon steel / galvanized / stainless steel option
Process piping	Carbon steel / 304SS / 316LSS / cryogenic SS / project-specific
Meter wetted parts	Selected to product and Coriolis model
Strainer body	CS / SS
Air eliminator	CS / SS; product-specific internals
Valves	CS / SS / cryogenic construction as applicable
Gaskets / seals	Product and temperature compatible
Instrument tubing	316SS typical
Fasteners	Project coating / stainless option

Available Options

- Duplex filtration; finer product filtration; sampling point; proving/calibration connections.
- Density / temperature compensation and standard-volume calculation where required.
- Additive injection package.
- Top, bottom or side loading arm interface.
- Vapour return / recovery connection.
- Printer / ticketing / RFID / vehicle identification interface.
- TAS / PLC / SCADA integration, remote I/O and historian interface.
- Heating, insulation, tracing and recirculation for viscous products.
- Emergency release coupling / breakaway for LPG/LNG applications.

8. Ordering Code - Proposed

LNS-MS / SERVICE / SIZE / METER / CONTROL / LOADING / OPTIONS

Field	Examples	Meaning
SERVICE	LH / FO / LPG / LNG	Product-family variant
SIZE	Project DN / NPS	Nominal skid process line size
METER	CMF	Coriolis mass flowmeter
CONTROL	EX-BC / TAS	Explosion-proof batch controller or TAS-integrated control
LOADING	TOP / BOTTOM / SIDE	Loading interface
EARTHING	E	Earthing
VAPOUR RECOVERY	V	Vapour Recovery
EARTHING	E	Earthing
OVERFILL	O	Overfill Prevention system
OPTIONS	H / A / P	Heating, Additive, Proving etc.

9. Project Data Required for Final Selection

Required Input	Project Entry
Product / composition	TBD
Minimum / normal / maximum flow	TBD
Batch size / tanker capacity	TBD
Design / operating pressure	TBD
Design / operating temperature	TBD
Density / viscosity / vapour pressure	TBD
Line size / flange rating	TBD
Hazardous-area classification	TBD
Required metering accuracy / custody status	TBD
Loading method and arm/hose interface	TBD
Power supply / instrument air	TBD
Communication / TAS / PLC interface	TBD
Material / corrosion requirements	TBD
Required codes, inspections and certification	TBD

10. Document Notes

This technical datasheet is a product-family document. Values identified as project-specific or TBD shall be completed on the project-specific LNS-MS-LPG datasheet before equipment selection, sizing or manufacture.

